

DEPLOYMENT PLAN

Learnly Educational Platform

Project Information

Project Name: Learnly - Educational Platform

Team Name: True Learners

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1. DEPLOYMENT OVERVIEW

1.1 System Architecture

Learnly platformu üç ana bileşenden oluşmaktadır:

- Backend:** Spring Boot 3.5.7 (Java 17) - REST API
- Frontend:** React 18 + Vite - Single Page Application
- Database:** MySQL 8.0 - Relational Database

1.2 Technology Stack

Backend Technologies

- Spring Boot 3.5.7
- Spring Security + JWT Authentication
- Spring Data JPA (Hibernate)
- Maven Build Tool

Frontend Technologies

- React 18

- Vite (Build Tool)
- Axios (HTTP Client)
- React Router (Navigation)

Database

- MySQL 8.0
- JDBC Connection

1.3 Deployment Environment

- **Development Mode:** Local development environment
- **Backend Port:** 8080
- **Frontend Port:** 5173
- **Database Port:** 3306

2. DEPLOYMENT PROCESS

2.1 Prerequisites

```
# Required software
- Java JDK 17
- Node.js 18+
- MySQL 8.0
- Maven 3.8+
- Git
```

2.2 Database Setup

Step 1: Create Database

```
-- MySQL'e bağlan
mysql -u root -p

-- Database oluştur
CREATE DATABASE learny_db CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;

-- Kullanıcı oluştur (opsiyonel)
CREATE USER 'learny_user'@'localhost' IDENTIFIED BY 'secure_password';
GRANT ALL PRIVILEGES ON learny_db.* TO 'learny_user'@'localhost';
FLUSH PRIVILEGES;
```

Step 2: Tables Auto-Created

Tablolar JPA tarafından otomatik oluşturulur:

- Kullanıcılar (Users)
- Dersler (Courses)
- Sınavlar (Exams)
- Sorular (Questions)
- Seçenekler (Options)
- ÖğrenciSınavSonuçları (Exam Results)
- KayıtlıÖğrenciler (Enrollments)

2.3 Backend Deployment

Step 1: Clone Repository

```
git clone <https://github.com/m-Gonul/True_Learners>
cd True_Learners
```

Step 2: Configure Database

src/main/resources/application.properties :

```
spring.datasource.url=jdbc:mysql://localhost:3306/learnny_db
spring.datasource.username=root
spring.datasource.password=yourpassword
spring.jpa.hibernate.ddl-auto=update
spring.jpa.show-sql=false
server.port=8080
```

Step 3: Build and Run

```
# Dependencies yükle
mvn clean install

# Uygulamayı başlat
mvn spring-boot:run

# Alternatif: JAR oluştur ve çalıştır
mvn clean package
java -jar target/Learnny-0.0.1-SNAPSHOT.jar
```

Step 4: Verify Backend

```
# Health check
curl http://localhost:8080/api/health

# Test endpoint
curl http://localhost:8080/api/courses
```

2.4 Frontend Deployment

Step 1: Navigate to Frontend

```
cd smart-education_dashboard
```

Step 2: Install Dependencies

```
npm install
```

Step 3: Configure Proxy

vite.config.js (zaten yapılandırılmış):

```
export default defineConfig({
  plugins: [react()],
  server: {
    proxy: {
      '/api': {
        target: 'http://localhost:8080',
        changeOrigin: true,
      },
    },
  },
})
```

Step 4: Run Development Server

```
npm run dev
```

Step 5: Verify Frontend

Browser'da aç: `http://localhost:5173`

2.5 Complete Startup Sequence

```
# Terminal 1: MySQL başlat
sudo service mysql start
```

```
# Terminal 2: Backend başlat
cd LearnY
mvn spring-boot:run
```

```
# Terminal 3: Frontend başlat
cd smart-education_dashboard
npm run dev
```

3. CONFIGURATION PLAN

3.1 Backend Configuration

application.properties

```
# Database Configuration
spring.datasource.url=jdbc:mysql://localhost:3306/learnny_db
spring.datasource.username=root
spring.datasource.password=yourpassword
spring.datasource.driver-class-name=com.mysql.cj.jdbc.Driver

# JPA/Hibernate Configuration
spring.jpa.hibernate.ddl-auto=update
spring.jpa.show-sql=false
spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.MySQLDialect
spring.jpa.properties.hibernate.format_sql=true

# Server Configuration
server.port=8080
server.error.include-message=always

# Logging Configuration
logging.level.root=INFO
logging.level.com.True_Learners.Learnny=DEBUG
```

JWT Configuration (JwtUtil.java)

```
private static final String SECRET_KEY = "your-256-bit-secret-key-here";
private static final long EXPIRATION_TIME = 86400000; // 24 saat
```

Security Configuration

```
// CORS: Tüm origin'lere izin (development)
@Bean
public CorsConfigurationSource corsConfigurationSource() {
    CorsConfiguration configuration = new CorsConfiguration();
    configuration.setAllowedOrigins(Arrays.asList("*"));
    configuration.setAllowedMethods(Arrays.asList("GET", "POST", "PUT", "DELETE"));
    configuration.setAllowedHeaders(Arrays.asList("*"));
    return source;
}

// Public endpoints
.requestMatchers("/api/auth/**").permitAll()
.requestMatchers(HttpMethod.GET, "/api/courses/**").permitAll()
```

```
.requestMatchers(HttpMethod.GET, "/api/exams/**").permitAll()
```

```
// Protected endpoints  
.anyRequest().authenticated()
```

3.2 Frontend Configuration

package.json

```
{  
  "name": "smart-education-dashboard",  
  "version": "1.0.0",  
  "type": "module",  
  "scripts": {  
    "dev": "vite",  
    "build": "vite build",  
    "preview": "vite preview"  
  },  
  "dependencies": {  
    "react": "^18.2.0",  
    "react-dom": "^18.2.0",  
    "react-router-dom": "^6.x",  
    "axios": "^1.x"  
  }  
}
```

API Configuration (axios)

```
// Base URL  
const API_BASE_URL = 'http://localhost:8080/api';  
  
// Axios interceptor (JWT token)  
axios.interceptors.request.use(config => {  
  const token = localStorage.getItem('token');  
  if (token) {  
    config.headers.Authorization = `Bearer ${token}`;  
  }  
  return config;  
});
```

3.3 Environment Variables

Development

```
# Backend  
SPRING_PROFILES_ACTIVE=dev  
DB_HOST=localhost  
DB_PORT=3306
```

```
DB_NAME=learny_db
```

```
# Frontend
```

```
VITE_API_URL=http://localhost:8080
```

Production (Future)

```
# Backend
```

```
SPRING_PROFILES_ACTIVE=prod
```

```
DB_HOST=production-db-host
```

```
JWT_SECRET=production-secret-key
```

```
# Frontend
```

```
VITE_API_URL=https://api.learny.com
```

3.4 Database Schema

Auto-Generated Tables

```
-- Users table
```

```
CREATE TABLE Kullanicilar (  
    Kullaniciid INT PRIMARY KEY AUTO_INCREMENT,  
    Kullaniciadi VARCHAR(50) UNIQUE,  
    Sifrehash VARCHAR(255),  
    Adsoyad VARCHAR(100),  
    Mail VARCHAR(100) UNIQUE,  
    Rol ENUM('STUDENT', 'TEACHER')  
);
```

```
-- Courses table
```

```
CREATE TABLE Dersler (  
    Dersid INT PRIMARY KEY AUTO_INCREMENT,  
    Dersadi VARCHAR(100),  
    Aciklama TEXT,  
    Ogretmenid INT,  
    FOREIGN KEY (Ogretmenid) REFERENCES Kullanicilar(Kullaniciid)  
);
```

```
-- Exams table
```

```
CREATE TABLE Sinavlar (  
    Sinavid INT PRIMARY KEY AUTO_INCREMENT,  
    Dersid INT,  
    Baslik VARCHAR(150),  
    Aciklama TEXT,  
    Suredakika INT,  
    Olusturmatarihi DATETIME,  
    FOREIGN KEY (Dersid) REFERENCES Dersler(Dersid)  
);
```

```
-- Additional tables: Sorular, Secenekler, OgrenciSinavSonuclari, etc.
```

4. TASK MATRIX

Task	Team Member	Description
Database Setup	Tüm Ekip	MySQL kurulumu ve schema oluşturma
Backend Configuration	Tüm Ekip	Spring Boot configuration ve dependency management
Security Setup	Tüm Ekip	JWT authentication ve authorization
Frontend Setup	Tüm Ekip	React + Vite configuration
API Integration	Tüm Ekip	Frontend-Backend bağlantısı
Testing	Tüm Ekip	Deployment verification
Documentation	Tüm Ekip	Deployment dokümanı hazırlama

DEPLOYMENT CHECKLIST

- MySQL database oluşturuldu
- Backend dependencies yüklendi
- application.properties yapılandırıldı
- JWT security ayarlandı
- Backend başarıyla başlatıldı (port 8080)
- Frontend dependencies yüklendi
- Vite proxy yapılandırıldı
- Frontend başarıyla başlatıldı (port 5173)
- API endpoints test edildi
- Authentication flow doğrulandı